

LAB EXPERIMENT RECORD

SET 12: WEBSITE TRAFFIC ANALYSIS

Department	AIDS
Course Code / Course Name	DSA0602 / Data Handling and Visualization
Experiment	Data Visualization using R and Tableau
Name	B KIRTHANAA
Register Number	192324244
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Day	Tuesday

Aim

To analyze website traffic data using a line chart, a bar chart and a data table in R, and to build an interactive dashboard using Tableau. Dataset: Website Traffic

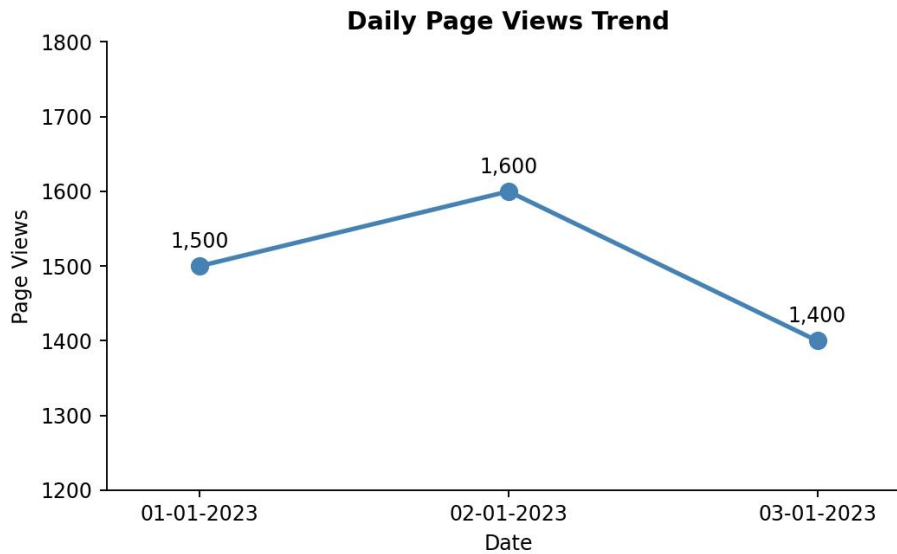
Date	Page Views	Click-through Rate
2023-01-01	1500	2.3%
2023-01-02	1600	2.7%
2023-01-03	1400	2.0%

PROGRAM 1: Line Chart of Daily Page Views

Create a line chart to visualize the trend in daily page views over time. Label the axes and title the chart.

```
traffic <- data.frame(
  Date      = as.Date(c("2023-01-01", "2023-01-02", "2023-01-03")),
  Page_Views = c(1500, 1600, 1400),
  CTR_Percent = c(2.3, 2.7, 2.0)
)

plot(traffic$Date, traffic$Page_Views,
     type = "o",
     col  = "steelblue",
     lwd  = 2,
     pch  = 16,
     xaxt = "n",
     xlab = "Date",
     ylab = "Page Views",
     main = "Daily Page Views Trend",
     ylim = c(1200, 1800))
axis.Date(1, at = traffic$Date, format = "%d-%m-%Y")
text(traffic$Date, traffic$Page_Views, labels = traffic$Page_Views, pos = 3)
```

OUTPUT:**Output 1: Daily Page Views Trend (Line Chart)**

Observation: Page views rise from 1,500 on 1 January to 1,600 on 2 January (about 6.7%) and then fall to 1,400 on 3 January (about 12.5%). 2 January is the peak day and 3 January is the lowest. The total for the three days is 4,500 page views, an average of 1,500 per day.

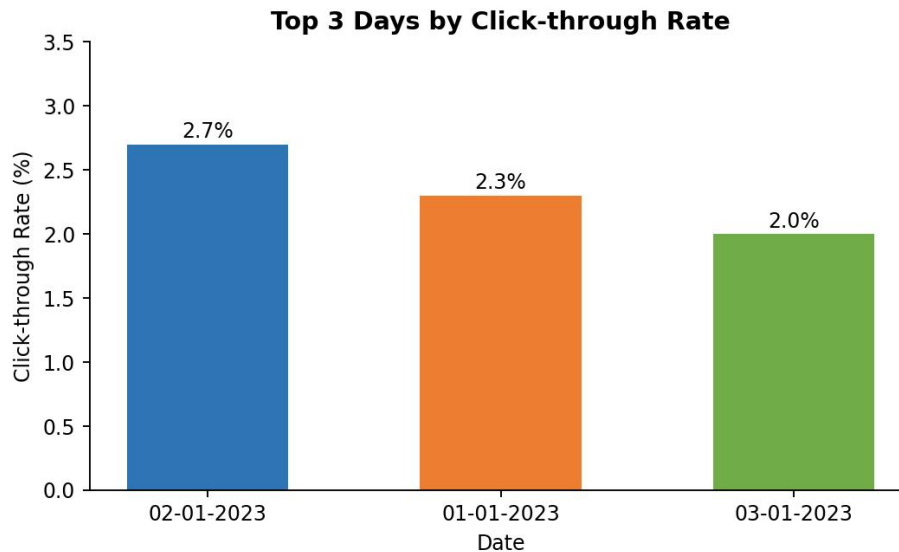
PROGRAM 2: Bar Chart of Top N Days by Click-through Rate

Generate a bar chart showing the top N days with the highest click-through rates. Label the chart elements.

```
N <- 3

top_days <- head(traffic[order(-traffic$CTR_Percent), ], N)

bp <- barplot(top_days$CTR_Percent,
              names.arg = format(top_days$Date, "%d-%m-%Y"),
              col = c("steelblue", "darkorange", "forestgreen"),
              xlab = "Date",
              ylab = "Click-through Rate (%)",
              main = paste("Top", N, "Days by Click-through Rate"),
              ylim = c(0, 3.5))
text(bp, top_days$CTR_Percent, labels = paste0(top_days$CTR_Percent, "%"), pos = 3)
```

OUTPUT:**Output 2: Top 3 Days by Click-through Rate (Bar Chart)****Observation:**

2 January has the highest click-through rate at 2.7%, followed by 1 January at 2.3% and 3 January at 2.0%. The day with the most page views (2 January) also has the best click-through rate, while the day with the fewest page views (3 January) has the lowest.

PROGRAM 3: Table of Website Traffic for Each Day

Build a table to show the website traffic data for each day. Label the table elements.

```
traffic <- data.frame(
  Date      = as.Date(c("2023-01-01", "2023-01-02", "2023-01-03")),
  Page_Views = c(1500, 1600, 1400),
  CTR_Percent = c(2.3, 2.7, 2.0)
)

print(traffic)

sum(traffic$Page_Views)
mean(traffic$CTR_Percent)
```

OUTPUT:**Console Output**

```
   Date Page_Views CTR_Percent
1 2023-01-01     1500         2.3
2 2023-01-02     1600         2.7
3 2023-01-03     1400         2.0

[1] 4500
[1] 2.333333
```

Output 3a: Website Traffic Table (R Console)

Date	Page Views	Click-through Rate (%)
2023-01-01	1500	2.3
2023-01-02	1600	2.7
2023-01-03	1400	2.0
Total / Average	4500	2.33

Output 3b: Website Traffic Table with Total and Average

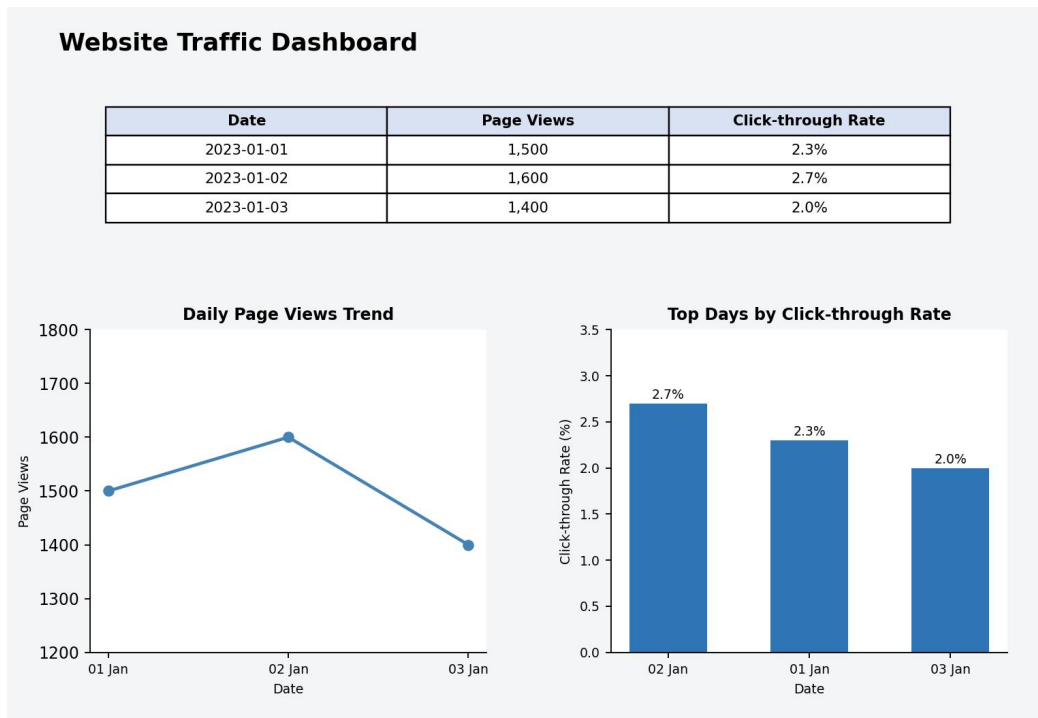
Observation: The table gives the exact page views and click-through rate for each day. The three days together received 4,500 page views with an average click-through rate of about 2.33%. 2 January was the strongest day on both measures and 3 January the weakest.

PROGRAM 4: Interactive Website Traffic Dashboard in Tableau

Develop a Tableau dashboard combining the line chart, bar chart and the table for interactive exploration of website traffic data.

Steps in Tableau

1. Save the data as an Excel or CSV file and connect to it in Tableau (Connect > Microsoft Excel / Text file). Check that Date is a Date field, and convert Click-through Rate to a number (for example 2.3) if it is read as text.
2. Sheet 1 (Line Chart): Drag Date to Columns (choose Day, exact date) and Page Views to Rows (SUM). Add Page Views to Label and give the axes and title meaningful names. Rename the sheet "Page Views Trend".
3. Sheet 2 (Bar Chart): Drag Date to Columns and Click-through Rate to Rows (AVG). Right-click Date > Filter > Top tab > By field, choose Top, create a parameter N, and rank by Average of Click-through Rate. Sort descending, add Click-through Rate to Label and rename the sheet "Top CTR Days".
4. Sheet 3 (Table): Drag Date to Rows, Measure Names to Columns and Measure Values to Text. Keep only Page Views and Click-through Rate in Measure Values. Rename the sheet "Traffic Table".
5. Create a new Dashboard (Dashboard > New Dashboard), set the size to Automatic, and drag the three sheets onto the dashboard area.
6. Add interactivity: select each chart and click Use as Filter, so clicking a day in one view filters the others. Add a Date filter and the N parameter control (Show Filter > Apply to Worksheets > All Using This Data Source).
7. Add the title "Website Traffic Dashboard" and test the filters in Presentation Mode.

OUTPUT:**Output 4: Interactive Website Traffic Dashboard (Tableau)****Observation:**

The dashboard shows the daily page view trend, the days with the highest click-through rates and the traffic table together. Selecting a day in any view filters the other two.

Procedure

1. Define the website traffic data in R as a data frame with Date, Page_Views and CTR_Percent.
2. Use plot() with type = "o" to create the line chart of daily page views and label the axes and title.
3. Order the data by click-through rate with order(), select the top N rows with head(), and draw them with barplot().
4. Display the traffic data with print() and calculate the total page views and average click-through rate with sum() and mean().
5. Load the traffic file into Tableau and build the line chart, bar chart and table worksheets.
6. Combine the worksheets in a Tableau dashboard and add filters and actions for interactivity.

RESULT:

The website traffic data was successfully analyzed using a line chart, a bar chart and a data table in R. An interactive Tableau dashboard was also created, combining all three views so that users can explore the daily page views and click-through rates.

Key Functions and Tools Used

Function / Tool	Purpose
<code>data.frame()</code> / <code>as.Date()</code>	Creates the traffic table and converts text to dates.
<code>plot(type = "o")</code>	Creates the line chart of daily page views.
<code>order()</code> / <code>head()</code>	Sorts by click-through rate and selects the top N days.
<code>barplot()</code>	Creates the bar chart of top N days.
<code>print()</code> / <code>sum()</code> / <code>mean()</code>	Displays the table and calculates total and average values.
Tableau Top N Filter	Limits the bar chart to the top N days by click-through rate.
Tableau Dashboard	Combines worksheets with filters and actions.